

Paint Protection Film (PPF)

Core Course — 3-Day Beginner Program

Student Workbook · Version 1.0

Student Name: _____

Class Date: _____

Instructor: _____

Location: _____

Paint Protection Film (PPF) — Student Workbook

Skill Forge Core Course · 3-Day Beginner Program

Welcome

Welcome to the Skill Forge Paint Protection Film (PPF) Core Course. Over three intensive days you will learn the foundational skills to install automotive paint protection film — from surface preparation and your first flat-panel install to completing a professional partial-front package under supervision.

PPF is one of the highest-value services in automotive restyling. This course gives you the hands-on practice and quality standards to start offering PPF with confidence.

What "Core" means: This course is designed for beginners with little or no PPF experience. Vinyl wrap or window tint experience helps but is not required. By Day 3 you will have completed multiple panel installs and a supervised partial-front package. Advanced techniques (full vehicle wraps, color PPF, complex bumpers) are covered in our Advanced courses.

Pre-Course Checklist

Before You Arrive

- ☐ Wear closed-toe shoes and shop-appropriate clothing all three days
- ☐ Bring safety glasses (loaners available)
- ☐ Remove rings, watches, and bracelets that can scratch paint or film
- ☐ Trim fingernails — you will handle adhesive film constantly
- ☐ If you have tint or wrap experience, bring one question you want answered about PPF differences

Provided Each Day

- Practice panels and vehicle access for supervised installs
 - PPF film (practice and project material)
 - Tools: squeegees, slip solution, heat gun, blades, tack cloths, prep chemicals
 - Lunch and refreshments
-

Course Schedule Overview

Day	Theme	Focus
Day 1	Foundations	Film science, paint assessment, tools, prep, flat panels
Day 2	Technique	Stretching, edges, hood, bumpers, mirrors, complex curves

DAY 1

Foundations

Day 1 Learning Objectives

By the end of Day 1, you will be able to:

1. Identify PPF film types and explain TPU vs. older PVC constructions
 2. Assess paint condition and determine if correction is needed before install
 3. Set up a PPF workstation with contamination control
 4. Perform a complete paint decontamination and prep sequence
 5. Install film on a flat practice panel with clean edges and no lift
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Day 1 Schedule

Time	Session
8:00 – 8:30	Welcome, introductions, course overview
8:30 – 9:30	Classroom: PPF film types, construction & self-healing
9:30 – 10:15	Classroom: Paint assessment & correction basics
10:15 – 10:30	Break
10:30 – 11:30	Classroom: Tools, slip solution, workspace setup
11:30 – 12:00	Classroom: Safety — blades, chemicals, heat, paint damage
12:00 – 1:00	Lunch
1:00 – 1:45	Demo: Full decon and prep sequence on hood section
1:45 – 3:15	Lab 1: Flat panel install — hood section
3:15 – 3:30	Break
3:30 – 4:45	Lab 2: Edge work — wrap, tuck, and trim on practice panel
4:45 – 5:00	Day 1 knowledge check & Q&A

Session 1: PPF Film Types & Construction

What Is Paint Protection Film?

PPF is a thick, optically clear (or colored/textured) polyurethane film applied to painted surfaces to protect against rock chips, scratches, bug acids, bird droppings, and UV damage. Modern films are **TPU (thermoplastic polyurethane)** — flexible, self-healing, and durable.

Film Construction

Layer	Function
Top coat	Self-healing, stain resistance, hydrophobic properties
TPU body	Impact absorption, flexibility, clarity
Adhesive	Pressure-sensitive; designed for clean removal without paint damage
Release liner	Protects adhesive until install

Film Categories

Type	Description	Common Use
Standard gloss	Crystal clear, high gloss	Full front, high-impact zones
Matte / satin	Reduces gloss of factory paint	Matte wraps, stealth packages
Colored PPF	Pigmented TPU	Color-change protection
Textured	Carbon fiber look, etc.	Accent panels, trim



PRO TIP

Installer tip: Self-healing works with heat — light scratches disappear with sun exposure or warm water. Explain this to customers; it's a major selling point for premium films.

Self-Healing Explained

The top coat is an elastomeric polymer that flows back into place when heated above ~140°F. This is **not** magic — deep gouges that penetrate the TPU body cannot self-heal. Set customer expectations accurately.

Session 2: Paint Assessment & Correction Basics

Before Any PPF Install, Inspect Paint

Walk the vehicle in good lighting (preferably outdoors or with LED inspection lights):

- ☐ Rock chips and deep scratches — document pre-existing damage
- ☐ Swirl marks and oxidation — may need polish before film
- ☐ Repainted panels — adhesion can vary; test or disclose risk
- ☐ Contamination — tar, tree sap, rail dust, iron particles
- ☐ PPF-unfriendly surfaces — fresh paint (cure time), vinyl wraps, plasti-dip

Paint Correction Decision

Condition	Action
Light swirls, film will hide minor defects	Wash, decon, alcohol wipe — proceed
Moderate swirls visible through film	Single-stage polish recommended
Deep scratches, chips to primer	Document; fill chips if needed; disclose to customer
Fresh paint (< 30 days)	Do not install — outgas will cause adhesive failure



SAFETY

Always photograph pre-existing damage. PPF makes paint look better — customers may blame you for damage that was already there.

Decontamination Sequence (Standard)

1. **Wash** — two-bucket method, clean mitts, blow out emblems and gaps
2. **Clay bar or clay mitt** — remove bonded contamination with lubricant
3. **Iron remover** — dissolve embedded ferrous particles (purple reaction = working)
4. **Tar/sap remover** — spot treat as needed
5. **Isopropyl alcohol wipe (70% IPA)** — remove oils and residue; final prep
6. **Inspect under light** — any remaining particles must be removed before film

Session 3: Tools, Slip Solution & Workspace

Essential PPF Tools

Tool	Purpose
Hard squeegee (with felt or low-friction sleeve)	Primary wet-out tool
Soft squeegee	Final passes, delicate areas
Slip/tack solution bottles	Positioning and final adhesion
Heat gun	Stretching, conforming, post-heating
Snitty knife / plotter blade	Trimming film edges
Extension blade holder	Precise edge trimming in tight areas
Tack cloth / lint roller	Remove dust before film contact
Plotter (shop)	Pre-cut patterns — introduced Day 3
Light bar / inspection lamp	Reveal contamination and defects

Slip vs. Tack Solution

Solution	Purpose	When
Slip (soapy water)	Allows film to slide for positioning	During placement and initial squeegee
Tack (higher alcohol or dedicated tack solution)	Increases adhesion for final tack	Edges, wraps, and problem areas

Follow your film manufacturer's recommended ratios. Too much slip = film won't stick. Too much tack too early = can't reposition.

Workspace Requirements

- Temperature: 65–75°F ideal for adhesive performance
- Dust control: avoid active sanding nearby; wet-mop floors
- Lighting: bright, angled light reveals every particle
- Organized film: rolls stored flat or on rack; handle edges carefully

Session 4: Safety



SAFETY

Non-negotiable safety rules:

1. Blades cut paint and skin — use fresh blades; cut away from body
2. Heat guns reach extreme temperatures — never hold stationary on paint or film
3. Chemical PPE: gloves and eye protection when using IPA, iron remover, and adhesives
4. Never install on a hot panel in direct sun — adhesive behaves unpredictably
5. Lift panels properly — hoods and doors are heavy; use props and assistance

Lab 1: Flat Panel Install — Hood Section



HANDS-ON LAB

Lab 1 — Hood Section Install

Objective: Prep a hood section and install a piece of PPF with clean wet-out and no trapped contamination.

Time: 90 minutes

Part A — Decon & Prep (25 min)

1. Wash and clay the hood section
2. Iron remover if needed; rinse and dry
3. IPA wipe the entire install zone
4. Final inspection under light — zero particles allowed
5. **Checkpoint:** Panel squeaks clean under your hand

Part B — Film Handling & Position (20 min)

1. Cut film from roll with 1–2" margin beyond panel edges
2. Peel liner partially; spray adhesive side with slip solution
3. Spray panel with slip solution
4. Position film; align with body lines or panel edges
5. **Checkpoint:** Film slides freely for final positioning

Part C — Wet-Out (35 min)

1. Anchor the center — light tack squeegee pass in the middle
2. Work outward in overlapping strokes; push solution to edges
3. Use firm, even pressure — PPF is thicker than tint
4. Post-heat lightly if minor fingers remain (instructor guidance)
5. **Checkpoint:** No bubbles, no visible contamination, film fully adhered

Part D — Self-Assessment

Edge cleanliness ___ | Bubble count ___ | Contamination ___ | Overall (1–5) ___

Lab 2: Edge Work — Wrap, Tuck & Trim



HANDS-ON LAB

Lab 2 — Edge Techniques

Objective: Practice the three primary edge finishing methods on a practice panel.

Time: 75 minutes

Edge Method 1 — Cut to Edge (Flush)

- Film trimmed exactly at panel edge
- Clean look; no wrap
- Requires precise cutting; edge is vulnerable to lift over time

Edge Method 2 — Wrap Edge

- Film wraps 2–3mm around panel edge to backside
- More durable; preferred on hoods, fenders, doors
- Use heat to conform; squeegee wrap zone firmly

Edge Method 3 — Tuck into Gap

- Film tucked into panel gaps (hood-to-fender, door jambs)
- Invisible edge; premium finish
- Requires gap access and precise trimming with extension blade

Practice all three methods on your assigned panel sections.

Instructor sign-off: ☐ Pass ☐ Redo

Day 1 Knowledge Check



KNOWLEDGE CHECK

1. What does TPU stand for, and why is it preferred over older PVC films?
2. How does self-healing work, and what are its limitations?
3. List the six steps of a standard decontamination sequence.
4. Why should you never install PPF on fresh paint?
5. What is the difference between slip solution and tack solution?
6. Name three edge finishing methods and when each is appropriate.

Day 1 Notes

DAY 2

Technique Development

Day 2 Learning Objectives

By the end of Day 2, you will be able to:

1. Apply controlled stretch to conform film to moderate compound curves
2. Use relief cuts appropriately without compromising coverage
3. Complete a full hood install with wrapped edges
4. Install film on a bumper section and mirror cap
5. Identify and correct common defects: fingers, silvering, contamination, stretch marks

Day 2 Schedule

Time	Session
8:00 – 8:30	Day 1 review & Q&A
8:30 – 9:45	Classroom: Stretching, anchor points & relief cuts
9:45 – 10:30	Demo: Hood install from bulk roll
10:30 – 10:45	Break
10:45 – 11:45	Classroom: Bumpers, mirrors & complex curves
11:45 – 12:00	Classroom: Troubleshooting — fingers, silvering, lift
12:00 – 1:00	Lunch
1:00 – 2:45	Lab 3: Full hood install (supervised)
2:45 – 3:00	Break
3:00 – 4:30	Lab 4: Bumper section + mirror cap install
4:30 – 5:00	Day 2 knowledge check & Q&A

Session 5: Stretching, Anchors & Relief Cuts

Controlled Stretch

PPF can stretch 10–40% depending on film brand and technique. **Controlled stretch** means applying tension only where needed to conform to curves — not stretching the entire panel uniformly.

Anchor Point Strategy

1. **Identify the flattest zone** — anchor here first (center of hood, flat bumper section)
2. **Work toward curves** — stretch incrementally as you squeegee outward
3. **Heat assists stretch** — warm film conforms without excessive force
4. **Never stretch beyond manufacturer limit** — film thins, adhesives stress, silvering appears

Relief Cuts

A relief cut is a carefully placed slit in excess film (not in the installed area) that allows the film to open and conform around sharp compound curves, mirror bases, and badge areas.

Rules for relief cuts:

- Cut in **excess margin only** — never in the visible install zone
- Cut from the edge inward — stops before the adhered area
- One cut at a time — reassess after each
- Seal cut edges with heat and tack solution after conforming



SAFETY

Silvering (whitish haze along adhesive) means the film was overstretched or adhesive was stressed. It may not disappear. Remove and replace the section.

Session 6: Bumpers, Mirrors & Complex Curves

Bumper Strategy

Bumpers are the most complex PPF install zone — compound curves, sensor cutouts, and tight edges.

1. **Pattern vs. bulk** — beginners should use plotter patterns when available
2. **Work from center outward** — same anchor principle as hood
3. **Heat frequently** — short bursts, keep moving
4. **Sensor and camera cutouts** — punch or trim precisely; don't cover ADAS sensors unless customer waives (not recommended)
5. **Relief cuts are normal** on aggressive bumper curves — hide them in inconspicuous zones

Mirror Caps

1. Remove cap if possible — flat install on bench is easier
2. If on-vehicle: stretch from center; heavy heat on compound curves
3. Wrap edges for durability — mirror caps see abrasion
4. Post-heat entire cap after install

Session 7: Troubleshooting Common Defects

Defect	Likely Cause	Fix
Finger	Insufficient stretch/heat on curve	Lift, reheat, re-squeegee; or remove and redo
Silvering	Overstretch or adhesive stress	Remove section; install fresh film
Contamination	Skipped decon step or dusty shop	Remove film, re-prep, reinstall
Edge lift	Incomplete tack on wrap zone	Tack solution + firm squeegee + post-heat
Stretch marks	Film pulled too hard	Remove and replace — marks don't heal
Orange peel in adhesive	Trapped solution	Squeegee harder; post-heat; severe cases = redo

Lab 3: Full Hood Install



HANDS-ON LAB

Lab 3 — Full Hood Install

Objective: Install PPF on a full hood from bulk roll with wrapped edges and no visible defects.

Time: 105 minutes

Vehicle assigned: _____

Film used: _____

Edge method: ☐ Wrap ☐ Tuck ☐ Flush

Instructor sign-off: ☐ Pass ☐ Redo

Lab 4: Bumper Section & Mirror Cap



HANDS-ON LAB

Lab 4 — Bumper & Mirror Cap Install

Objective: Install PPF on a bumper section and mirror cap, using relief cuts if needed.

Time: 90 minutes

Bumper section: _____

Relief cuts used: ☐ Yes ☐ No — Location: _____

Mirror cap: ☐ On-vehicle ☐ Off-vehicle

Instructor sign-off: ☐ Pass ☐ Redo

Day 2 Knowledge Check



KNOWLEDGE CHECK

1. What is an anchor point, and how do you choose one?
2. When is a relief cut appropriate, and where should it never be placed?
3. What causes silvering, and can it be fixed?
4. Why are bumper installs considered more difficult than hood installs?
5. Describe the post-heat process and why it matters.
6. What is the maximum stretch approach — stretch everything or stretch only where needed?

Day 2 Notes

DAY 3

Professional Practice

Day 3 Learning Objectives

By the end of Day 3, you will be able to:

1. Explain common PPF packages (partial front, full front, track, full body)
2. Use pattern/plotter software at a beginner level (or follow pre-cut patterns)
3. Complete a supervised partial-front package to shop quality standards
4. Perform a QC inspection using a standardized checklist
5. Outline pricing structure for PPF services

Day 3 Schedule

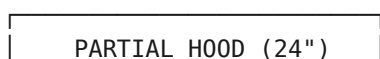
Time	Session
8:00 – 8:30	Day 2 review & Q&A
8:30 – 9:30	Classroom: PPF packages & coverage zones
9:30 – 10:15	Classroom: Pattern software & bulk vs. plotter workflow
10:15 – 10:30	Break
10:30 – 11:00	Classroom: QC inspection & customer delivery
11:00 – 12:00	Demo: Partial front package walkthrough
12:00 – 1:00	Lunch
1:00 – 1:45	Classroom: Pricing, upselling, and business basics
1:45 – 4:15	Lab 5: Partial front package — capstone install
4:15 – 4:45	Final knowledge check
4:45 – 5:00	Course wrap-up, certificates, feedback

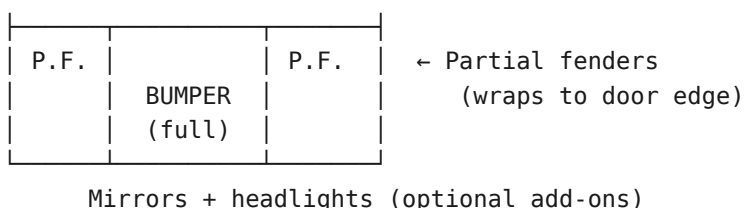
Session 8: PPF Packages & Coverage Zones

Common Packages

Package	Typical Coverage	Target Customer
High-wear	Headlights, mirrors, door cups, rocker panels	Budget-conscious; high-impact zones
Partial front	Partial hood, partial fenders, full bumper, mirrors	Most popular entry package
Full front	Full hood, full fenders, bumper, mirrors, headlights	Premium daily-driver protection
Track package	Full front + rocker panels, A-pillars, rear splash	Enthusiast / track use
Full body	Entire painted surface	Collector, exotic, long-term ownership

Coverage Zone Diagram (Partial Front)





PRO TIP

Upsell script: "The partial front covers where 80% of rock chips happen. Full front adds complete hood and fender coverage — most customers who plan to keep the car 3+ years go full front."

Session 9: Pattern Software & Workflow

Bulk Roll vs. Plotter Pattern

Method	Pros	Cons
Bulk roll	No software cost; full control	More waste; slower; harder on complex panels
Plotter pattern	Precise fit; faster; less waste	Software subscription; database gaps for new models

Beginner Workflow (Plotter)

1. Select vehicle year/make/model/trim in pattern software
2. Choose kit (partial front, full hood, etc.)
3. Nest patterns on roll for material efficiency
4. Plot/cut on machine
5. Weed liner from cut pieces
6. Install each piece in logical order: center panels first, then edges, then details

Pattern Software at Your Shop

Your shop will use one plotter pattern platform. Learn that system deeply — vehicle lookup, kit selection, nesting, and cut workflow. Ask your instructor which platform you will train on and where to find vehicle patterns for new models.

Session 10: QC Inspection & Customer Delivery

QC Checklist — Every Install

- ☐ Paint prepped — decon and IPA complete; pre-existing damage documented
- ☐ No contamination under film (inspect with light at angle)
- ☐ No fingers, silvering, or stretch marks in visible areas
- ☐ Edges wrapped, tucked, or flushed per spec — no lift
- ☐ All plotter pieces aligned — no visible seams misaligned at panel lines
- ☐ Emblems, badges, and sensors properly handled

- ☐ ADAS sensors uncovered and functional
- ☐ Post-heat completed on all installed panels
- ☐ Customer paperwork with package details and warranty info

Customer Delivery

1. Walk the vehicle with the customer in good lighting
2. Point out coverage zones and any documented pre-existing paint damage
3. Explain cure time: **no wash for 48 hours**; hand wash only for first week
4. Provide written aftercare: no pressure washer directly on edges, no abrasive compounds on film
5. Explain warranty: film warranty (manufacturer) vs. workmanship warranty (your shop)

Session 11: Pricing & Business Basics

Pricing Components

Component	Notes
Material cost	Film cost per sq ft × coverage area + 15–20% waste factor
Labor	Hours × shop rate (PPF labor rate is higher than tint)
Complexity premium	Exotics, tight bumpers, color PPF
Correction add-on	Polish/decon if not included in base

Sample Package Pricing Framework

Package	Typical Retail Range*
High-wear	\$300 – \$600
Partial front	\$800 – \$1,400
Full front	\$1,500 – \$2,500
Full body	\$5,000 – \$10,000+

**Varies significantly by market, film brand, and vehicle type. Research your local market.*

Profitability Tips

- **Film efficiency** — nest patterns tightly; track waste per job
- **Bundle services** — PPF + ceramic coating on film is a strong upsell
- **Deposit policy** — 50% deposit on PPF bookings; material cost is high
- **Photography** — before/after photos are your best marketing asset

Lab 5: Partial Front Package — Capstone Install



HANDS-ON LAB

Lab 5 — Partial Front Package (Capstone)

Objective: Complete a supervised partial-front PPF package meeting all QC checklist items.

Time: 150 minutes

Vehicle assigned: _____

Package includes: ☐ Partial hood ☐ Partial fenders ☐ Full bumper ☐ Mirrors ☐ Headlights

Film brand & thickness: _____

Install method: ☐ Plotter pattern ☐ Bulk roll

QC checklist completed: ☐ Yes

Instructor final sign-off: ☐ Pass ☐ Redo

Final Knowledge Check



KNOWLEDGE CHECK

1. What panels are typically included in a partial front package?
2. What is the difference between bulk roll and plotter pattern installs?
3. List five items on the PPF QC checklist.
4. How long should a customer wait before washing their vehicle?
5. Name three components of PPF pricing.
6. What is silvering, and how do you prevent it?



CERTIFICATE OF COMPLETION

Paint Protection Film (PPF)

Core Course — 3-Day Beginner Program

This certifies that

successfully completed all course requirements including
labs, knowledge checks, and the capstone install.

Requirement	Completed	Instructor
Day 1 Labs	☐	_____
Day 2 Labs	☐	_____
Day 3 Capstone	☐	_____
All Knowledge Checks	☐	_____

Class Date

Instructor

Issue Date



Scan for Skill Forge Resources

Course updates, reference materials, and
future Skill Forge programs.

forgeppf.com/skill-forge

Instructor Signature

Skill Forge Program
Director

Resources & Quick Reference

Decon Sequence

Wash → Clay → Iron remover → Tar spot treat → IPA wipe → Inspect

Stretch Rule

Anchor flat zone first → stretch only where needed → heat assist → post-heat all panels

Edge Priority

Tuck (best appearance) → Wrap (best durability) → Flush (acceptable on hidden edges)

Aftercare

48 hrs no wash · Hand wash first week · No pressure on edges · Annual inspection recommended

